# lab4

資料工碩一 313551027 曹咏萱

## iperf UDP

`iperf -c 10.6.1.2 -u -t 30`

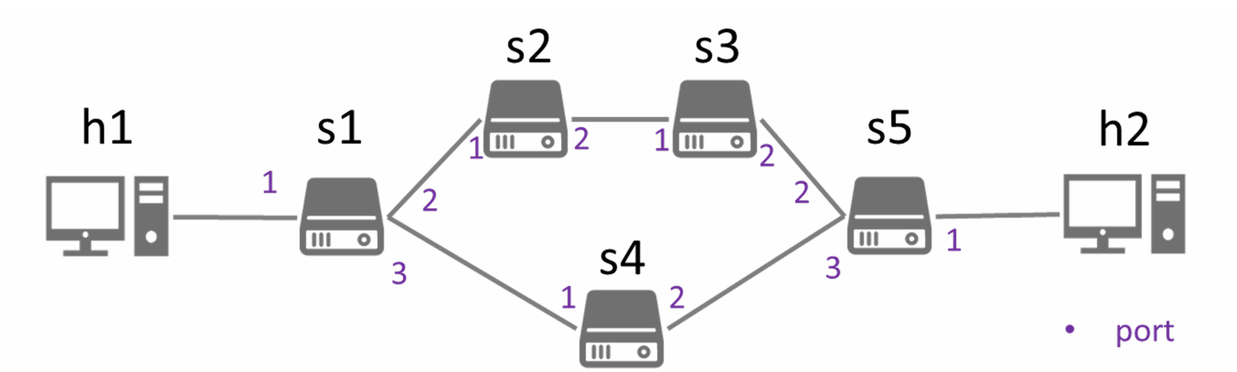
```
root@SDN-NFV:/home/reki/SDN/Lab4# sudo iperf -c 10.6.1.2 -u -t 30
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 54572 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-30.0121 sec  3.75 MBytes  1.05 Mbits/sec
[  1] Sent 2679 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-30.0118 sec  3.75 MBytes  1.05 Mbits/sec   0.005 ms 0/2678 (0%)
root@SDN-NFV:/home/reki/SDN/Lab4#
```

`sh ovs-ofctl dump-ports –O OpenFlow14 s1`

```
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s1
OFPST_PORT reply (OF1.4) (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=441.667s
  port  "s1-eth1": rx pkts=10734, bytes=8286250, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=291, bytes=39609, drop=0, errs=0, coll=0
           duration=441.679s
  port  "s1-eth2": rx pkts=285, bytes=38925, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=11003, bytes=8323939, drop=0, errs=0, coll=0
           duration=441.679s
  port  "s1-eth3": rx pkts=288, bytes=39435, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=286, bytes=38995, drop=0, errs=0, coll=0
           duration=441.679s
```

```
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s4
OFPST_PORT reply (OF1.4) (xid=0x2): 3 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=445.792s
  port  "s4-eth1": rx pkts=288, bytes=39273, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=290, bytes=39713, drop=0, errs=0, coll=0
           duration=445.812s
  port  "s4-eth2": rx pkts=290, bytes=39713, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=285, bytes=38925, drop=0, errs=0, coll=0
           duration=445.812s
mininet>
```

> **RX/TX packets**: Number of received/transmitted packets.

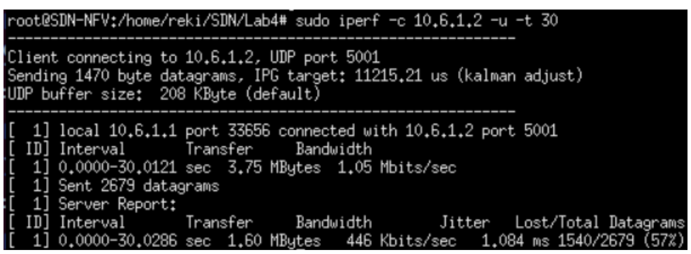



Packets transmitted by port 2 are much more than from port 3, which means traffic from h1 to h2 will go through s2 and s3.

s4 port 2 will receive packets from h2 and send them to h1.

`sh ovs-ofctl dump-ports -O OpenFlow14 s4`

## link s1 s2 down



```
root@SDN-NFV:/home/reki/SDN/Lab4# sudo iperf -c 10.6.1.2 -u -t 30
------------------------------------------------------------
Client connecting to 10.6.1.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.6.1.1 port 33656 connected with 10.6.1.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-30.0121 sec  3.75 MBytes  1.05 Mbits/sec
[  1] Sent 2679 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-30.0286 sec  1.60 MBytes   446 Kbits/sec   1.084 ms 1540/2679 (57%)
```

After turning down link s1 s2, bandwidth will be limited, and 57% of packets were dropped.

```
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s1
OFPST_PORT reply (OF1.4) (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=317.772s
  port  "s1-eth1": rx pkts=60, bytes=35248, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=185, bytes=24875, drop=0, errs=0, coll=0
           duration=317.791s
  port  "s1-eth2": rx pkts=153, bytes=20577, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=187, bytes=46859, drop=0, errs=0, coll=0
           duration=317.792s
  port  "s1-eth3": rx pkts=182, bytes=24701, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=187, bytes=31643, drop=0, errs=0, coll=0
           duration=317.791s
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s1
OFPST_PORT reply (OF1.4) (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=349.819s
  port  "s1-eth1": rx pkts=1851, bytes=1418960, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=207, bytes=27867, drop=0, errs=0, coll=0
           duration=349.838s
  port  "s1-eth2": rx pkts=153, bytes=20577, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=187, bytes=46859, drop=0, errs=0, coll=0
           duration=349.839s
  port  "s1-eth3": rx pkts=203, bytes=27651, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=1997, bytes=1418093, drop=0, errs=0, coll=0
           duration=349.838s
```

```
mininet> sh ovs-ofctl dump-ports -O OpenFlow14 s4
OFPST_PORT reply (OF1.4) (xid=0x2): 3 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=1438.071s
  port  "s4-eth1": rx pkts=11649, bytes=8413531, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=991, bytes=161397, drop=0, errs=0, coll=0
           duration=1438.091s
  port  "s4-eth2": rx pkts=991, bytes=161397, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=8589, bytes=4100619, drop=0, errs=0, coll=0
           duration=1438.091s
```

`sh ovs-ofctl dump-ports –O OpenFlow14 s1` before and after link-down s1 s2 can tell form that since port2's packets didn't gain, the traffic went through port3 to s4.